Lab 3: Docker

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ITI _SYSTEM_ADMIN 46

ALEXANDRIA

1.part 1:

Run an insecure docker registry (cncf distribution) on your server, follow the docs for this https://docs.docker.com/registry/insecure/
Create an image that installs and run nginx based on alpine:latest
Push this created image to the private registry that you created in step 1

1. **Start the registry container:**

$ docker run -d -p 5000:5000 --restart=always --name registry registry:2

2. **Configure Docker to trust the local registry: Edit (or create) the /etc/docker/daemon.json file on your host:**

```
{
  "insecure-registries" : ["localhost:5000"]
}
```

3. **Restart Docker to apply changes:**

$ sudo systemctl restart docker

Then
Create the Nginx Image (Alpine-based)
1.Create docker file :

```
FROM alpine:latest
RUN apk add --no-cache nginx
RUN mkdir -p /run/nginx
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

2.build the image:
$ docker build -t localhost:5000/my-nginx:v1 .

3. Push the Image to Your Private Registry
$docker push localhost:5000/my-nginx:v1

4. Verify the push:
$curl -X GET http://localhost:5000/v2/_catalog

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root@serverA:~

```
[root@serverA ~]# docker run -d -p 5000:5000 --restart=always --name registry registry:2
Unable to find image 'registry:2' locally
2: Pulling from library/registry
6d464ea18732: Pull complete
44cf07d57ee4: Pull complete
bbbdd6c6894b: Pull complete
3493bf46cdec: Pull complete
8e82f80af0de: Pull complete
b537bf6d1146: Download complete
32a76c78501f: Download complete
Digest: sha256:a3d8aaa63ed8681a604f1dea0aa03f100d5895b6a58ace528858a7b332415373
Status: Downloaded newer image for registry:2
11771df14ace60c44397b53ac5a383c03b0af4b84facd9561e8a701afb08ecd6
[root@serverA ~]# vim /etc/docker/daemon.json
[root@serverA ~]# cat /etc/docker/daemon.json
{
  "insecure-registries" : ["localhost:5000"]
}
[root@serverA ~]# systemctl restart docker
```

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root@serverA:~/lab3_docker

```
[root@serverA ~]# cd lab3_docker/
[root@serverA lab3_docker]# vim Dockerfile
[root@serverA lab3_docker]# cat Dockerfile
FROM alpine:latest
RUN apk add --no-cache nginx
RUN mkdir -p /run/nginx
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]

[root@serverA lab3_docker]# docker build -t localhost:5000/my-nginx:v1 .
[+] Building 4.1s (7/7) FINISHED                                   docker:default
 => [internal] load build definition from Dockerfile                          0.0s
 => => transferring dockerfile: 215B                                          0.0s
 => [internal] load metadata for docker.io/library/alpine:latest              0.1s
 => [internal] load .dockerignore                                             0.0s
 => => transferring context: 2B                                               0.0s
 => [1/3] FROM docker.io/library/alpine:latest@sha256:25109184c71bdad752c8312a86232  0.0s
 => => resolve docker.io/library/alpine:latest@sha256:25109184c71bdad752c8312a86232  0.0s
 => [2/3] RUN apk add --no-cache nginx                                        3.3s
 => [3/3] RUN mkdir -p /run/nginx                                             0.2s
 => exporting to image                                                        0.3s
 => => exporting layers                                                       0.2s
 => => exporting manifest sha256:f60f7f9d2d805dabc80d6fc5f51d09e6793134b4ea74d17ae4  0.0s
 => => exporting config sha256:c52eac874160ddbe56827ba35fec412d6683679e882278b5e38f  0.0s
 => => exporting attestation manifest sha256:7e75e83fe7979ae0f9e017247af173c2de4606  0.0s
 => => exporting manifest list sha256:3b6c7637a3d2b69364fb11ac3f1ca4cb20deaed49337d  0.0s
 => => naming to localhost:5000/my-nginx:v1                                   0.0s
 => => unpacking to localhost:5000/my-nginx:v1                                0.0s
[root@serverA lab3_docker]# docker push localhost:5000/my-nginx:v1
The push refers to repository [localhost:5000/my-nginx]
7ebc3a31ec3a: Pushed
4f4fb700ef54: Pushed
589002ba0eae: Pushed
7aa2078f8c59: Pushed
v1: digest: sha256:3b6c7637a3d2b69364fb11ac3f1ca4cb20deaed49337debc8577f308c3062115 size:
855
[root@serverA lab3_docker]# curl -X GET http://localhost:5000/v2/_catalog
{"repositories":["my-nginx"]}
[root@serverA lab3_docker]#
```

2.part 2:

```
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                          root@serverA:~/lab3_docker
[root@serverA lab3_docker]# cat docker-compose.yml
services:
  db:
    image: mysql:5.7
    container_name: mysql_db
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpassword
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wordpress_user
      MYSQL_PASSWORD: wordpress_password
    volumes:
      - ./data:/var/lib/mysql

  wordpress:
    depends_on:
      - db
    image: wordpress:latest
    container_name: wordpress_app
    restart: always
    ports:
      - "8080:80"
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wordpress_user
      WORDPRESS_DB_PASSWORD: wordpress_password
      WORDPRESS_DB_NAME: wordpress
```
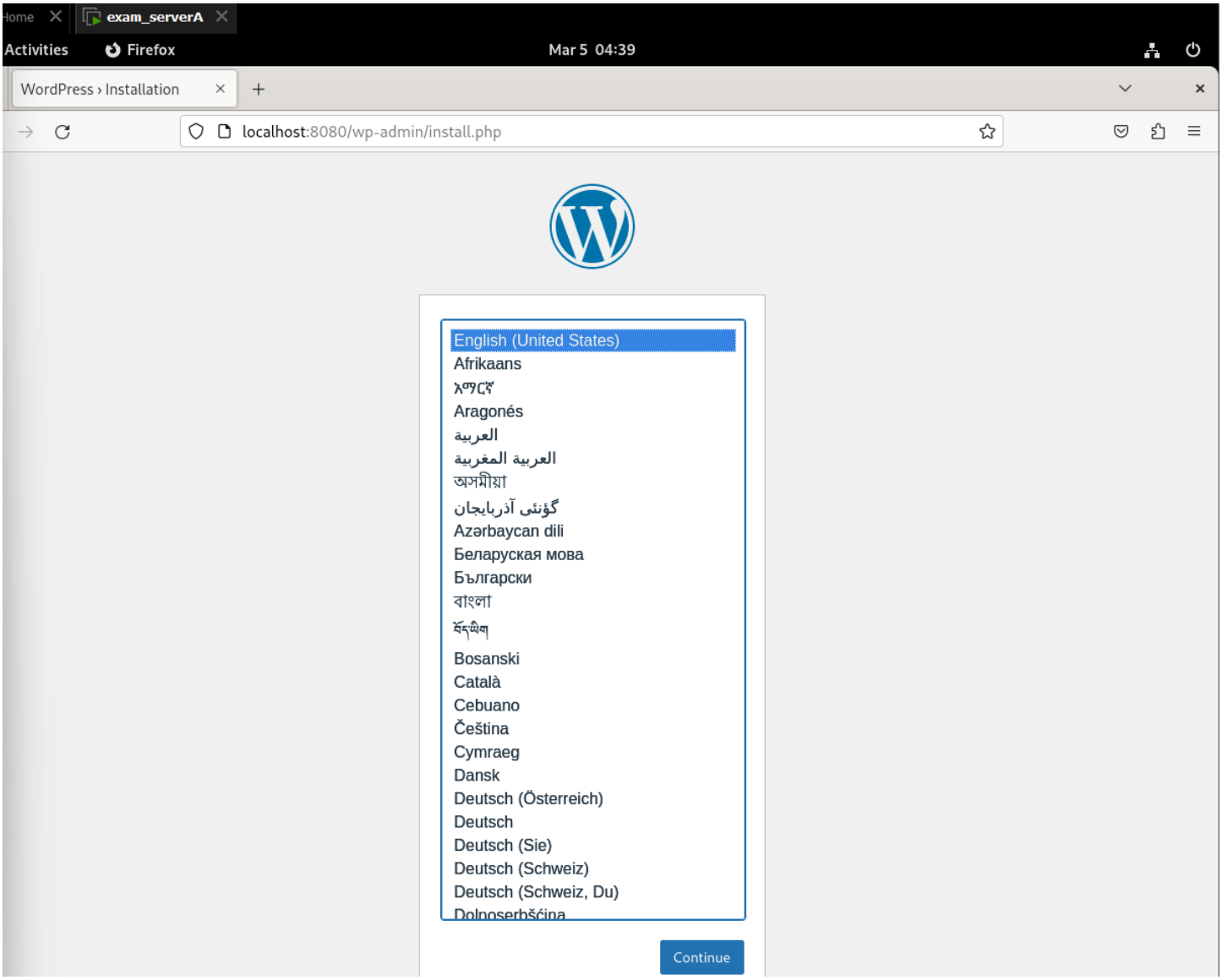

$ docker compose up -d

$ docker ps -a

Open your browser and go to http://localhost:8080 to finish the WordPress installation.



3.bonus-part:

```
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root@serverA:~/lab3_docker/part3
root@serverA:~/lab3_docker   root@serverA:~/lab3_docker/part3   root@serverA:~/lab3_docker/part3
[root@serverA part3]# cat  docker-compose.yml
version: '3'
services:
  flask-app:
    image: localhost:5000/iti-flask-lab2:latest
    container_name: flask-lab3
    restart: always

  nginx-app:
    image: nginx:alpine
    container_name: nginx-lab3
    restart: always
    ports:
      - "80:80"
    volumes:
      - ./nginx.conf:/etc/nginx/conf.d/default.conf
    depends_on:
      - flask-app

[root@serverA part3]# cat  nginx.conf
server {
    listen 80;

    location / {
        proxy_pass http://flask-app:5000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
[root@serverA part3]#
```

$ docker compose up -d

```
[root@serverA part3]# docker ps -a
CONTAINER ID   IMAGE                                   COMMAND                  CREATED
      STATUS              PORTS                                           NAMES
2fddaa6a3874   nginx:alpine                            "/docker-entrypoint.…"   6 minutes ago
      Up 6 minutes        0.0.0.0:80->80/tcp, [::]:80->80/tcp             nginx-lab3
ca361f65f9b5   localhost:5000/iti-flask-lab2:latest    "python app.py"          6 minutes ago
      Up 6 minutes        5000/tcp                                        flask-lab3
11771df14ace   registry:2                              "/entrypoint.sh /etc…"   About an hour
ago   Up About an hour    0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp     registry
[root@serverA part3]#
```

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Tiger Home Page

127.0.0.1

Tiger Home Page

Photo by John and Karen Hollingsworth, retouched by Zwoenitzer (Fish & Wildlife Service, ID WO0409-33F) [Public domain] via Wikimedia Commons

This is the index page for my example Flask app.

- In myth and legend
- As a symbol

The tiger (*Panthera tigris*) is the largest cat species, most recognisable for their pattern of dark vertical stripes on reddish-orange fur with a lighter underside. The species is classified in the